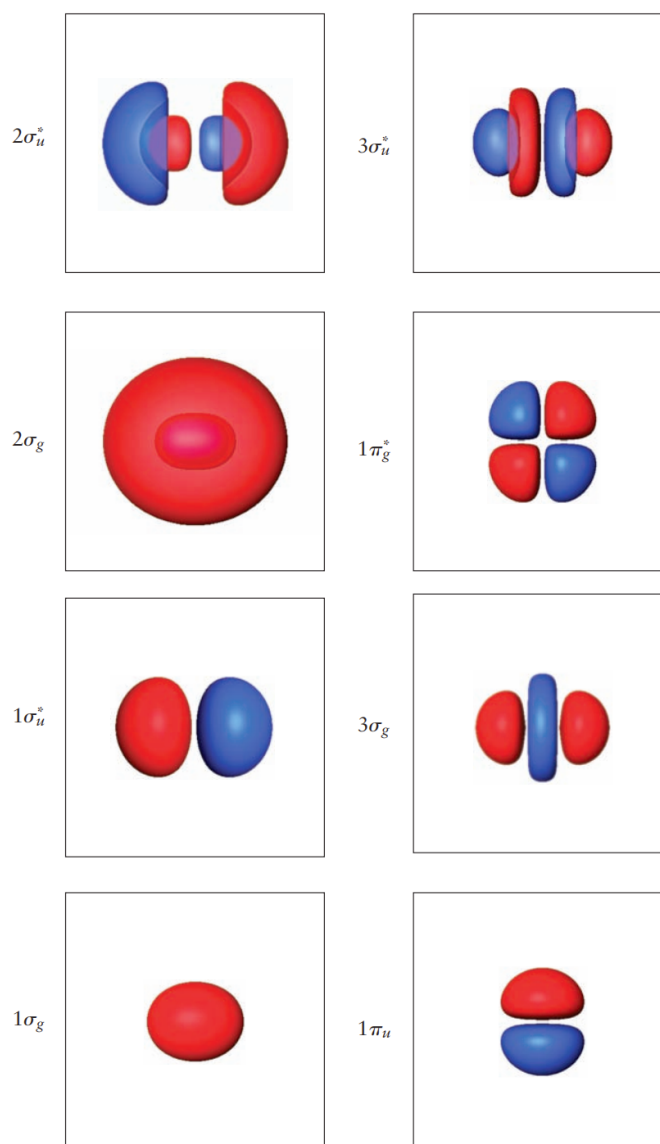


1. Determine the number of nodes along the internuclear axis and the number of nodal planes for each of the π molecular orbitals for H_2^+ shown in Figure 6.5.

FIGURE 6.5 Wave functions for the first eight energy levels of the H_2^+ molecular ion, calculated exactly by quantum mechanics. The two nuclei lie along the z-axis, which is in the plane of the paper. Regions of positive and negative values of the wave function are shown in red and blue, respectively. The labels for each orbital are explained in the text. The images are isosurfaces corresponding to contours at ± 0.1 of the maximum value. The $1\sigma_g$ is the orbital the electron occupies when H_2^+ is in its ground electronic state, and the energies increase in the order $1\sigma_g < 1\sigma_u^* < 2\sigma_g < 2\sigma_u^* < 1\pi_u < 3\sigma_g < 1\pi_g^* < 3\sigma_u^*$.
(Courtesy of Mr. Hatem Helal and Professor William A. Goddard III, California Institute of Technology, and Dr. Kelly P. Gauthier, University of Texas at Austin.)



2. Predict the valence electron configuration and the total bond order for the molecule I_2 . Will I_2 be paramagnetic or diamagnetic?
3. The compound nitrogen oxide (NO) forms when the nitrogen and oxygen in air are heated. Predict whether the nitrosyl ion (NO^+) will have a shorter or a longer bond than the NO molecule. Will NO^+ be paramagnetic like NO or diamagnetic?
4. The photoelectron spectrum of CO has four major peaks with ionization energies of 14.5, 17.2, 20.1, and 38.3 eV. Assign these peaks of molecular orbitals of CO, and prepare a quantitative energy level correlation diagram for CO. The ionization energy of carbon atoms is 11.26 eV, and the ionization energy of oxygen atoms is 13.62 eV.

5. Calcium carbide (CaC_2) is an intermediate in the manufacturing of acetylene (C_2H_2). It is the calcium salt of the carbide (also called acetylide) ion (C_2^{2-}). What is the electron configuration of this molecular ion? What is its bond order?